

Jiaming Wei | 魏佳铭

AI Evaluation / Red Teaming / Agent Reliability | Research Engineer

Beijing, China | jimmyenglish@126.com | quarkspace.top | GitHub | LinkedIn

EXPERIENCE

- Holistic AI** | Research Intern · LondonJun 2026 - Sep 2026
- Built **LLM / agent red-team and evaluation workflows** connecting attack generation, target execution, grading, and FP/FN analysis; focused on whether an apparent attack was a genuine breach or a measurement error.
 - Migrated existing red-team flows toward a more controllable, auditable evaluation pipeline integrated with the company platform; in multi-turn attacker-target-judge setups, separated **breach / refusal / provider filter / judge error** as distinct outcomes.
 - Audited graders against human/independent gold labels to identify systematic errors; related public methods and implementation were independently rebuilt in **redteam-under-test**.
- Huadian Group · Nanjing Nanzi Huadun Digital Technology** | AI Engineering InternAug 2025 - Sep 2025
- Worked on power-industry LLM and enterprise knowledge-system tasks, including domain adaptation, RAG/QA pipeline testing, and deployment-constraint analysis.

RESEARCH & PUBLICATIONS

- Routing Is Least Learnable Where It Is Most Valuable** | UCL MSc Dissertation2026
- Submitted to EMNLP 2026 Workshop REALM | NeurIPS 2026 Workshop VLM4RWD
- Designed a preregistered evaluation of Web-Agent page representations - DOM, Set-of-Marks, vision, and phantom variants - across **3 model families × 6 observation modes**.
 - Identified a **routing value-learnability gap**: representation portfolios retain real oracle headroom, yet learned routing is least reliable where routing would be most valuable; negative results and failure analysis form the main conclusion.
 - Built a **~117K-LOC Python / 1,121-test-function** research system spanning VisualWebArena, Playwright, heterogeneous-compute orchestration, structured logging/recovery, statistics, activation patching, and linear probes.

SELECTED SYSTEMS

- redteam-under-test** | Local-first LLM Red-Team Measurement Stack2026
- Mapped **166 plugins** to local generation mechanisms, enabled **103** after end-to-end verification, and reimplemented **11** attack techniques as local engines; attacks are instantiated conditionally on the target rather than replayed from a fixed prompt bank.
 - Treated the grader itself as a measurement instrument: against independent gold labels, the tuned judge achieved **.712-.753 recall at .93-.98 precision**, versus **.247 recall** for the upstream rubric on the same rows.
 - Built a zero-egress local execution path, cockpit, Postgres data layer, and local/cloud executors while preserving explicit breach/refusal/provider-filter/judge-error labels.
- Agent Red-Team Lab** | Tool-Using Agent Security / MCP2026
- Built a read-only, offline MCP tool-call trace analyzer for trust-boundary crossing, sensitive-data egress, destructive actions, confused-deputy behavior, and missing authorization.
 - Built a multi-step tool-attack replay framework that optimizes attack combinations under replay/model-hop budgets rather than prompt tuning alone; best public competition score: **72.72**.

EDUCATION

- University College London (UCL)** | MSc Artificial Intelligence for Sustainable Development · Computer ScienceSep 2025 - Sep 2026
- Xi'an Jiaotong University** | BEng AutomationSep 2021 - Jul 2025

SKILLS

- Evaluation / Research:** LLM & Agent Evaluation · Benchmark Design · Red Teaming · Judge/Grader Calibration · Failure Analysis · Bootstrap / Paired Testing
- Agents / Models:** Web & Computer-Use Agents · VisualWebArena · MCP · Tool Use · PyTorch · Hugging Face · LoRA/SFT · GRPO/RLVR
- Engineering:** Python · TypeScript/JavaScript · SQL/Postgres · Playwright · Docker · GitHub Actions · Linux · AWS/SST · Next.js/TanStack · SGE/HPC